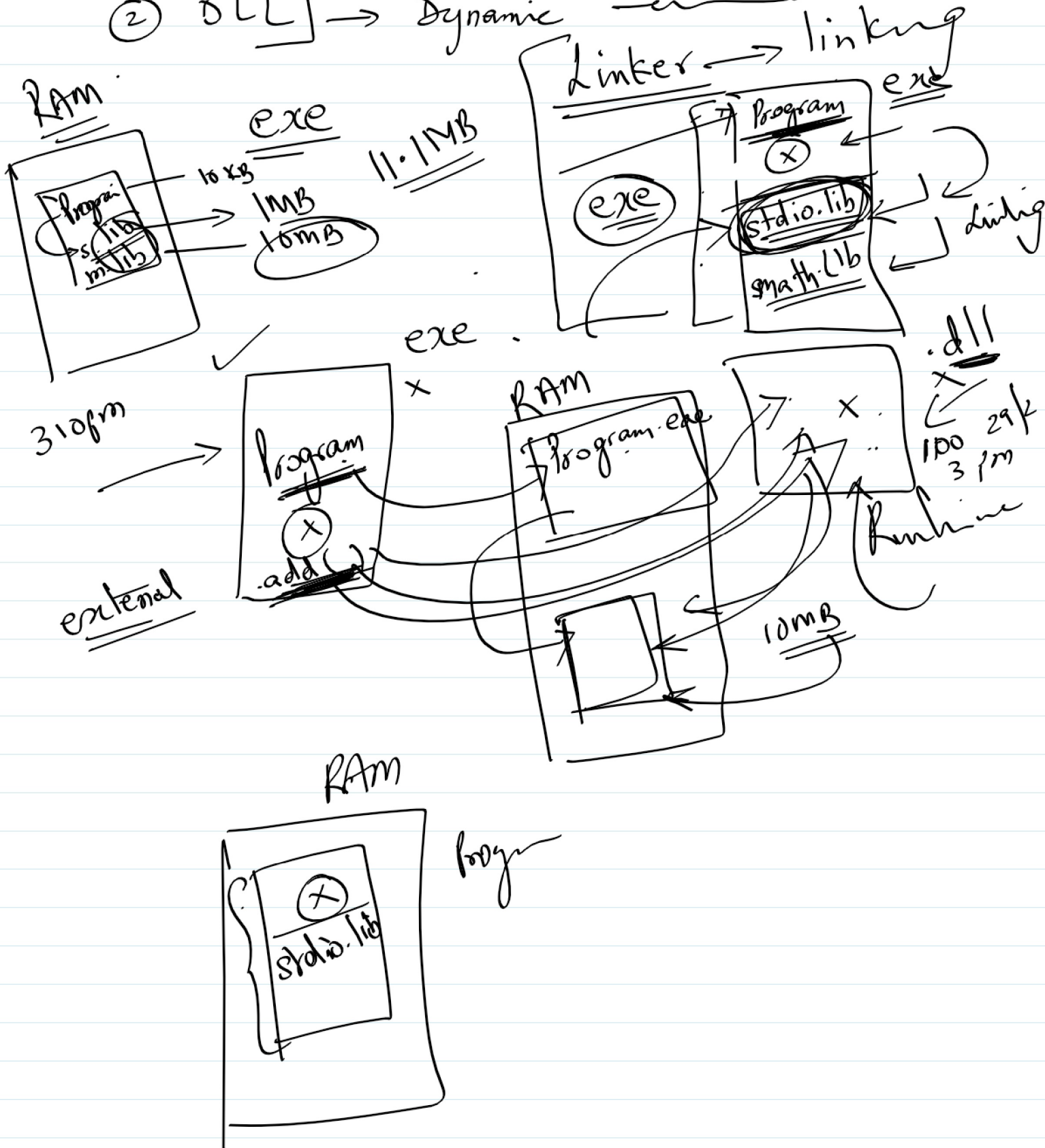


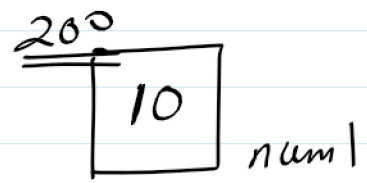
- h
- Binary
- ① SLL → Static linked Library
- ② DLL → Dynamic



```

void main()
{
    int num1;
    display(&num1);
}

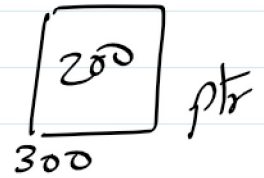
```



```

void display(int *ptr)
{
    *ptr = 10;
}

```



① Procedural $\rightarrow$ C

② Object Based $\rightarrow$ Basic

$\rightarrow$ ③ Object Oriented Language $\rightarrow$ Java C++
.NET,

$\rightarrow$ Object Real time Entity

- a) Identity
- b) State or Properties
- c) Behaviours / Methods

C $\rightarrow$ process-h
-beginthread ex

OOP.

Major

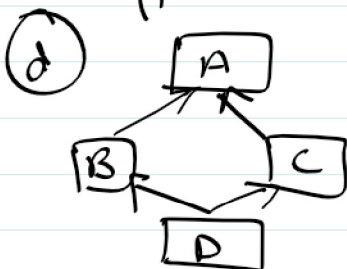
Minor

- ① Abstraction
- ② Encapsulation / Hiding
- ③ Inheritance
- ④ Polymorphism
- ⑤ Modularity

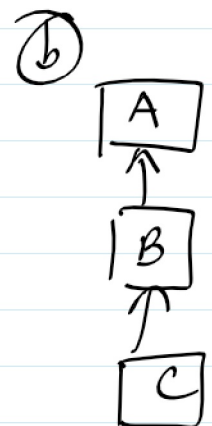
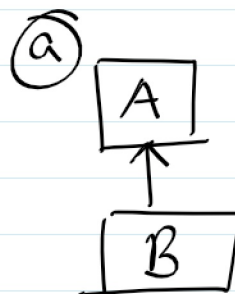
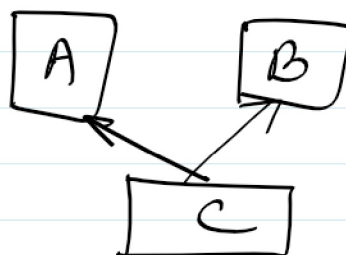
- ⑥ Persistence / Serialization
- ⑦ Concurrency?
- ⑧ RTTI / RTCI typeid
Runtime Type Identification
or
Runtime Class Information

③ Inheritance

- a) Single
- b) Multi level
- c) Multiple
- d) Hybrid



③



```

class A
{
    ~~~~~
    int data;
    void display()
    {
    }
}

class C : public A, B
{
    int value
}

```

```

class B
{
    ~~~~~
    int data;
    int num;
    void show()
    {
    }
}

void display()
{
}

```

C obj;

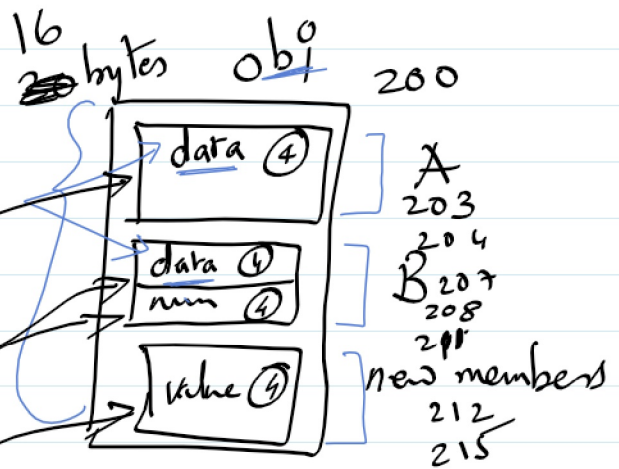
obj.data = 5;

obj.A::data = 5;

obj.B::data = 6;

obj.num = 7;

obj.value = 8;



④ Polymorphism

- Compile / Static / Early binding
- Runtime / Dynamic / Late binding

① Overloading

add@DIFI

add(int, float)

add(float, int)

add@FIDI

a function with same name but

- either the type of para change
- no. of para change
- sequence of data type of para sld change.

~~int add(int);~~
~~float add(int);~~

num = int
valr = float

```
printf("%.d", num);  
printf("%.f", value);  
printf("%.d %f", num, value);  
printf("%.f %.d", value, num);  
printf("%.d %.d", num, num);
```

```
int printf(const char r, ...);
```

↑ ellipses
variable argument
list

Name mangling → Compiler

C

function name

print(---);

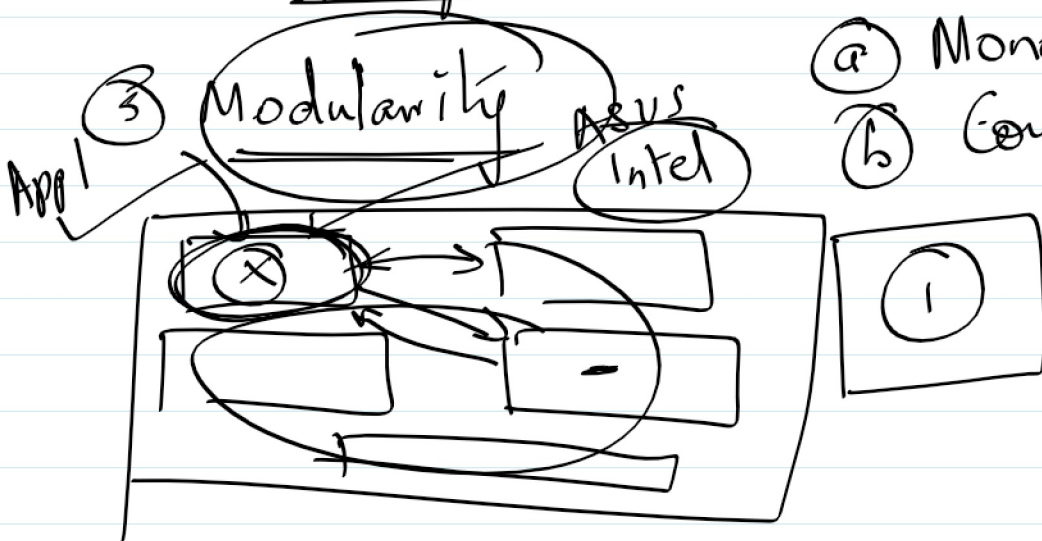
② print

C++

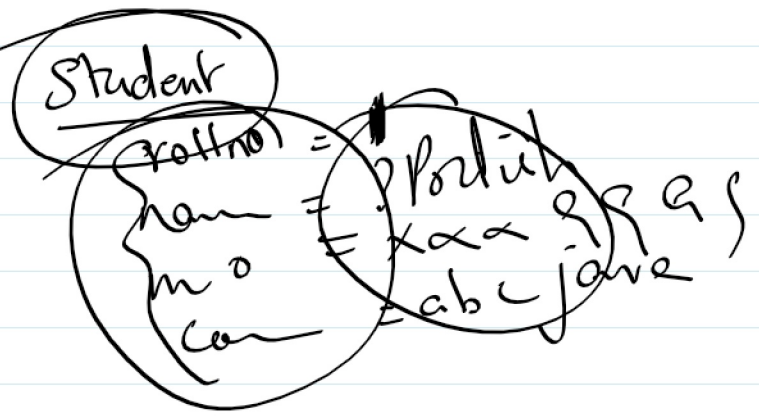
function name
+
parameters

(a) Monolithic Based Arch.
(b) Component Based Arch.

Ann 2 Lenovo
3 Dell
4 Intel
5



⑥ Persistence / Serialization
1, Prithish, Java, 99999



⑦ Concurrency

⑧ RTTI
RTCI